REGRESSION & SECURITY TEST REPORT

Student Attendance Management System (SAMS) | Issue #36 | TASK-005

1. Purpose and Scope

This report summarizes the recorded local checks for core SAMS workflows and selected security controls. Coverage includes authentication, authorization, session handling, student and academic setup, attendance operations, reporting, input validation, password storage, and basic recovery/performance smoke checks.

Test environment: Local FastAPI backend, static frontend, and SQLite test database. Test records are synthetic.

2. Execution Summary

Area	Recorded result	Notes
Functional workflows	Expected responses observed for recorded checks	Login, class/student/faculty setup, attendance marking and correction, history, reports, and CSV export.
Authorization and sessions	Restrictions and token invalidation observed	Student access to the admin audit endpoint returned HTTP 403; logout and expiry rejected token reuse.
Selected security checks	Checks recorded	Salted PBKDF2-HMAC-SHA256 password hashing observed; injection-style search returned no unintended records for the tested input.
Recovery and performance smoke checks	Successful with limited scope	Backend restart and five concurrent OpenAPI requests returned HTTP 200. These checks do not constitute formal load testing.
Deployment security	Pending verification	HTTPS/TLS was unavailable on the local development server and must be confirmed in the deployment environment.

3. Recorded Observations

OBS-001 — Subject creation: The first request returned HTTP 500 during a transient SQLite lock; a retry succeeded with HTTP 200. Investigate if the issue recurs.

OBS-002 — Automated test warnings: pytest reported 2 passed and 18 warnings. Review the warnings as maintenance items.

OBS-003 — UI coverage: Desktop login/dashboard was observed. Mobile responsiveness has not been separately assessed.

OBS-004 — HTTPS/TLS: HTTPS was unavailable on the local development server. Deployment TLS verification remains outstanding.

OBS-005 — Logging review: Source inspection was limited to backend/main.py; this does not establish comprehensive log redaction.

OBS-006 — Performance/concurrency: The recorded checks are smoke checks and do not establish production load capacity.

4. Defect Summary

The recorded notes include one transient SQLite lock during subject creation and the observations above. No additional confirmed defects are included in the available test record. Outstanding verification items are coverage gaps, not confirmed defects.

5. Conclusion

Recorded local checks demonstrated core attendance workflows, role restrictions, attendance correction/audit behavior, reporting, input validation, password hashing, and session expiry. Deployment TLS, mobile responsiveness, comprehensive logging review, automated-test warnings, and formal performance testing remain follow-up items as applicable.

6. Follow-up Actions

1. Confirm HTTPS/TLS in the deployment environment. 2. Assess mobile responsiveness. 3. Review logging across application modules and deployment logs. 4. Review pytest warnings. 5. Investigate the SQLite lock if it recurs. 6. Perform formal load testing if required by acceptance criteria.